

2023 Ch H1 Q23

Section: Chemistry in Society

Topic: The Mole Concept

Question Summary

Which of the following gas samples has the same volume as 16.0 g of oxygen gas (O_2), measured at the same temperature and pressure?

A: 21.0 g CO

B: 44.0 g CO_2

C: 46.0 g NO_2

D: 46.0 g N_2O_4

Worked Solution

At the same temperature and pressure, equal volumes of gases contain equal numbers of molecules (Avogadro's law). Therefore, the key is to compare the number of moles of gas.

Moles of $\text{O}_2 = 16.0 \text{ g} \div 32.0 \text{ g mol}^{-1} = 0.50 \text{ mol}$.

Now check each option:

- A: $21.0 \text{ g CO} \div 28.0 \text{ g mol}^{-1} = 0.75 \text{ mol}$

→ not equal.

- B: $44.0 \text{ g CO}_2 \div 44.0 \text{ g mol}^{-1} = 1.00 \text{ mol}$

→ not equal.

- C: $46.0 \text{ g NO}_2 \div 46.0 \text{ g mol}^{-1} = 1.00 \text{ mol}$

→ not equal.

- D: $46.0 \text{ g N}_2\text{O}_4 \div 92.0 \text{ g mol}^{-1} = 0.50 \text{ mol}$

→ equal to O₂.

Final Answer

D — 46.0 g of N₂O₄

Revision Tips

- Equal volumes of gases at the same T and P contain the same number of moles (Avogadro's law).

- Convert mass → moles to compare gas volumes.

- Remember: Volume depends only on moles, not type of gas.